CS 577 Deep Learning
Homework 3: Attention and Transformer Forward

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Fall 2026

Part 0: Setup and Axes

All experiments were executed on a CPU with a fixed random seed of `577`. The notebook environment used the following versions:

- Python: `3.11.18`
- NumPy: `2.4.6`
- PyTorch: `2.15.0`
- Matplotlib: `3.11.2`
- Device: `CPU`

All 11 public tests passed successfully. These tests covered stable masked softmax, causal masking, scaled dot-product attention, head splitting and merging, multi-head self-attention and cross-attention, positional encodings, and Transformer block behavior.

The final test summary was:

```
Ran 11 tests in 0.047s

OK

Public-test return code: 0
```

No failures or errors were reported.

0.2 Axis and mask contract

The attention scores S, attention weights A, and output Y have the following shapes:

$$S : (B, H, T_q, T_k), \qquad A : (B, H, T_q, T_k), \qquad Y : (B, H, T_q, d_v).$$

Here, B is the batch size, H is the number of heads, T_q is the number of query tokens, T_k is the number of key tokens, d_k is the query/key feature size, and d_v is the value feature size.

We apply softmax along the last axis (T_k) because each query compares itself with all keys. Softmax gives a probability to each allowed key, and these probabilities add up to 1. If all keys are blocked, the probabilities are all zero.

The key mask first has shape (B, T_k). We reshape it to $(B, 1, 1, T_k)$ because each batch may have a different mask, but the same mask is used for all heads and queries within that batch.

NumPy broadcasting works by matching dimensions from right to left. A dimension of size 1 can expand to match the corresponding dimension in another array.

This allows the mask with shape $(B, 1, 1, T_k)$ to work with attention scores of shape (B, H, T_q, T_k).

Part 1: Hand Calculation

1.1 Unmasked attention

TODO

1.2 Causal attention

TODO

1.3 Scaling and stability

TODO

Part 2: Stable Masked Attention

2.1 Numerical stability and fully blocked rows

TODO

2.2 Hand-case agreement

TODO

2.3 Padding and causal masks

TODO

Part 3: Multi-Head Attention

3.1 Shape trace, attention maps, and outputs

Stage	Meaning	Shape
X	input features	TODO
XW_Q, XW_K, XW_V	projected features	TODO
Q, K, V	after splitting heads	TODO
S, A	scores and weights per head	TODO
Head outputs	weighted values per head	TODO
Merged outputs	concatenated head features	TODO
Y_{MHA}	after output projection	TODO

TODO

3.2 Independent PyTorch comparison

Case	Maximum output error	Maximum weight error
Unmasked self-attention	TODO	TODO
Causal self-attention	TODO	TODO
Rectangular cross-attention	TODO	TODO

TODO

3.3 Head count and head-output ablation

TODO

Part 4: Transformer Block

4.1 Shape trace and reference checks

TODO

4.2 Positions and reordering

TODO

4.3 Future perturbation and causal visibility

TODO

Part 5: Architecture Comparison

5.1 Encoder–decoder and decoder-only generation

TODO

Assistance and Generative-AI Disclosure

TODO